Supplementary Material for SE-RGCL: Robust Graph Contrastive Learning against Adversarial Traffic Attacks in IoV

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APPENDIX A
SE-RGCL HYPERPARAMETER SETTINGS

Table I summarizes the complete hyperparameter configuration used in SE-RGCL. The parameter settings are kept fixed across the main experiments unless otherwise specified. The memory-bank capacities and hard negative mining configuration are selected to balance representation diversity and computational cost. The LLR regularization parameter follows the setting adopted in [9], while the remaining hyperparameters are selected according to validation performance.

The STM and LTM capacities are set to 512 and 2048, respectively, with an 8:2 LTM-to-STM sampling ratio to preserve long-term representation diversity while retaining recent batch information. The top $M = 128$ negatives with the highest cosine similarity to each anchor are used for hard negative mining. The EMA coefficient $\eta = 0.9$ stabilizes the long-term statistics across batches, and the temperature parameter is set to $\tau = 0.1$. The final objective uses $\epsilon_1 = 1.0$, $\epsilon_2 = 0.5$, and $\epsilon_3 = 0.1$ for the contrastive, prototype-alignment, and orthogonality terms, respectively.

TABLE I. Hyperparameter Settings of SE-RGCL.

Parameter	**Value**
GAT layers	2
Attention heads	8 / 1
Hidden dimension	512
Pre-training epochs T	100
Batch size $\|B\|$	256
STM capacity S	512
LTM capacity L	2048
LTM:STM sampling ratio	8:2
Hard negatives M	128
LLR regularization λ	0.001
EMA coefficient η	0.9
Temperature τ	0.1
Loss weight ϵ_1	1.0
Loss weight ϵ_2	0.5
Loss weight ϵ_3	0.1

APPENDIX B
CLEAN-DATA PERFORMANCE EVALUATION

We compare SE-RGCL with a wide range of GNN baselines, from foundational architectures (GCN [1], GAT [2], GIN [3], GraphSAGE [4]) to more advanced models that handle deeper propagation (GCNII [5]), improve robustness (Grand [6]), and aggregate higher-order information (MixHop [7], H_2GCN [8]). The evaluation of all models is conducted on the NSL-KDD and UNSW-NB15 datasets without adding any data perturbation (*i.e.*, no adversarial attacks are implemented) to accurately measure their inherent performance under ideal conditions. As detailed in Table II and Table III, SE-RGCL consistently achieves strong performance across the main evaluation metrics.

TABLE II
Model Performance Evaluation on the NSL-KDD Dataset.

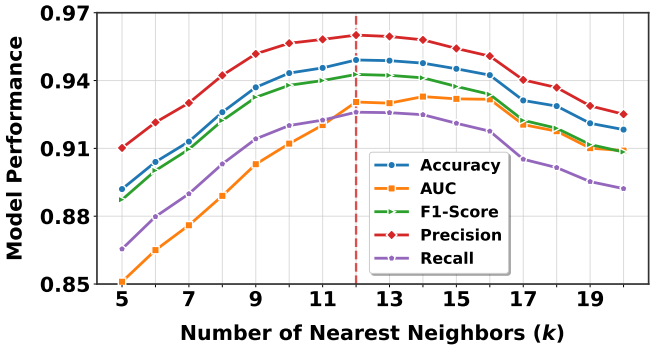
Models	Acc	Precision	Recall	F1 score	AUC
GCN	0.7650	0.9210	0.6430	0.7570	0.9150
GAT	0.7440	0.9090	0.6700	0.7710	0.7900
GIN	0.8060	0.9530	0.6940	0.8030	0.8240
GraphSAGE	0.8000	0.9650	0.6730	0.7930	0.8200
GCNII	0.8550	0.9701	0.7850	0.8651	0.9335
Grand	0.9091	0.9750	0.8010	0.8821	0.9512
MixHop	0.8750	0.9715	0.8552	0.9105	0.9400
H_2GCN	0.9527	0.9726	0.8234	0.8500	0.9601
SE-RGCL	**0.9821**	**0.9824**	**0.9818**	**0.9820**	**0.9820**

TABLE III
Model Performance Evaluation on the UNSW-NB15 Dataset.

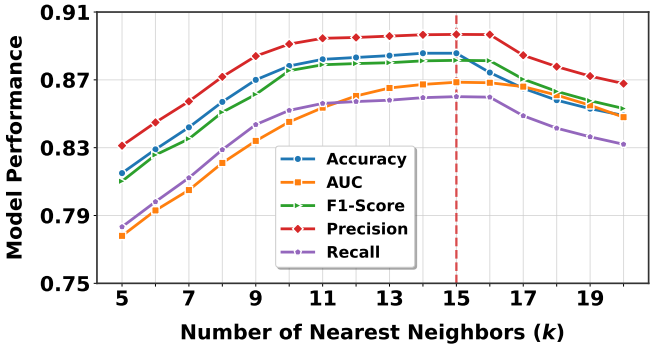
Models	Acc	Precision	Recall	F1 score	AUC
GCN	0.8446	0.8404	0.8395	0.8399	0.8904
GAT	0.9131	0.9101	0.9002	0.9047	0.9104
GIN	0.9154	0.9091	0.9071	0.9081	0.9015
GraphSAGE	0.9193	0.9167	0.9072	0.9116	0.9132
GCNII	0.9225	0.9280	0.9135	0.9205	0.9195
Grand	0.9310	0.9355	0.9108	0.9270	0.9260
MixHop	0.9216	0.9205	0.9255	0.9228	0.9295
H_2GCN	0.9132	0.9085	0.9211	0.9013	0.8926
SE-RGCL	**0.9477**	**0.9538**	**0.9289**	**0.9398**	**0.9336**

APPENDIX C
ADDITIONAL ANALYSIS OF NEIGHBORHOOD SIZE k

To further validate the structural entropy-guided neighborhood selection, we evaluate the IDS performance under different candidate values of k on NSL-KDD and UNSW-NB15. As shown in Fig. 1, the performance remains relatively stable around the neighborhood sizes selected by 1-DSE. For NSL-KDD, the highest performance is observed around $k = 12$, while the selected $k = 11$ lies within the corresponding high-performance region. For UNSW-NB15, the selected $k = 14$ is close to the empirical performance peak at $k = 15$. These results further support the use of 1-DSE for determining an appropriate neighborhood size for graph construction.



(a) Results on NSL-KDD dataset



(b) Results on UNSW-NB15 dataset

Fig. 1. Performance analysis of the IDS model with varying neighbor number k.

Appendix D
Performance Stability Analysis

To assess run-to-run variability, we repeated SE-RGCL five times with independent random seeds on NSL-KDD and UNSW-NB15 under clean conditions and the selected FGSM, MIM, and CW white-box attack settings, while keeping the data partitions and hyperparameters fixed. The random seeds affect model initialization, batch ordering, and, where applicable, adversarial-example generation. Table IV reports the mean and standard deviation of the F1 score and AUC across these five runs; the mean summarizes average performance, whereas the standard deviation describes inter-run variation.

FGSM is used during representation learning, whereas MIM and CW provide unseen iterative and optimization-based attack settings, respectively. SE-RGCL achieves high mean F1 scores and AUC values with small standard deviations across the five runs on both datasets. These results further support the strong and consistent detection performance of SE-RGCL across the five random seeds under the evaluated attack settings.

Appendix E
Sensitivity Analysis of Mobility-Aware Graph Parameters

We further analyze the sensitivity of SE-RGCL to the time-window length Δt, V2V communication radius R_{V2V}, and RSU communication radius R_{RSU} using a one-factor-at-a-time strategy. As shown in Table V, overly short or long time windows reduce performance due to insufficient temporal aggregation or weakened mobility consistency, respectively. Similarly, excessively small communication radii restrict feasible candidate connections, whereas overly large radii introduce less relevant node relationships. The settings $\Delta t = 10$s, $R_{\mathrm{V2V}} = 200$m, and $R_{\mathrm{RSU}} = 500$m provide the best overall performance on both datasets. The results demonstrate that SE-RGCL maintains stable performance over moderate parameter ranges and benefits from balanced temporal and communication constraints during graph construction.

TABLE IV. Performance Stability of SE-RGCL

Dataset	Setting	F1	AUC
NSL-KDD	Clean	0.9820 ± 0.0016	0.9820 ± 0.0014
	FGSM	0.9770 ± 0.0021	0.9765 ± 0.0018
	MIM	0.9775 ± 0.0023	0.9773 ± 0.0020
	CW	0.9064 ± 0.0047	0.9212 ± 0.0041
UNSW-NB15	Clean	0.9398 ± 0.0028	0.9336 ± 0.0025
	FGSM	0.9310 ± 0.0032	0.9301 ± 0.0029
	MIM	0.9307 ± 0.0035	0.9285 ± 0.0031
	CW	0.8890 ± 0.0056	0.8903 ± 0.0051

TABLE V. Sensitivity Analysis of Mobility-Aware Graph Parameters.

Parameter	Value	NSL-KDD		UNSW-NB15	
		F1	AUC	F1	AUC
Δt	5 s	0.9684	0.9701	0.9192	0.9236
	10 s	**0.9778**	**0.9776**	**0.9311**	**0.9320**
	15 s	0.9721	0.9734	0.9254	0.9271
	20 s	0.9647	0.9662	0.9168	0.9203
R_{V2V}	100 m	0.9689	0.9707	0.9201	0.9239
	200 m	**0.9778**	**0.9776**	**0.9311**	**0.9320**
	300 m	0.9734	0.9741	0.9276	0.9290
	400 m	0.9658	0.9673	0.9187	0.9214
R_{RSU}	300 m	0.9697	0.9712	0.9215	0.9247
	500 m	**0.9778**	**0.9776**	**0.9311**	**0.9320**
	750 m	0.9738	0.9746	0.9279	0.9293
	1000 m	0.9664	0.9681	0.9196	0.9221

Appendix F
Baseline Implementation and Hyperparameter Settings

For a fair comparison, all baseline methods are re-trained using the same training, validation, and test partitions, network traffic preprocessing procedures, and adversarial threat model as SE-RGCL. Model architectures and hyperparameters explicitly reported in the corresponding publications or official implementations are retained where applicable. Dataset-dependent or unspecified hyperparameters are selected according to validation performance, and the test set is used only for final evaluation after model and hyperparameter selection. The implementation and hyperparameter settings of all baseline methods are summarized in Table VI.

TABLE VI. Implementation and Hyperparameter Settings of Baseline Methods.

Method	Main Configuration	Key Settings / Parameter Selection
Artifact [10]	Bayesian uncertainty estimation using MC-dropout combined with KDE-based density estimation	MC-dropout $T = 50$; dropout rate $= 0.5$; KDE applied to deep-feature representations; original detection mechanism retained.
TAD [11]	Transfer-learning-based adversarial detector with three fully connected ReLU hidden layers	Three hidden layers with 256 units per layer; detector trained for 30 epochs; unspecified dataset-dependent settings selected according to validation performance.
Omni [12]	Hyperparameter-optimized ensemble of unexpected models with weighted ensemble prediction	TPE-based optimization over activation functions, hidden-layer sizes, dropout, optimizer, batch size, epochs, and learning rate; dropout $\in [0, 0.5]$; batch size $\in \{16, 32, 64, 128\}$; epochs $\in [5, 20]$; learning rate $\in \{0.001, 0.01, 0.1\}$.
SimGRACE [13]	Encoder-perturbation-based graph contrastive-learning framework	Batch size $= 128$; pre-training epochs $= 20$; original encoder-perturbation mechanism retained; dataset-dependent learning settings selected using the validation set.
DL Ens. [14]	Deep-learning ensemble with adversarial training, data balancing, feature preprocessing and engineering, and model fine-tuning	Original ensemble and adversarial-training strategy retained; method-specific unspecified settings selected according to validation performance.
RAIDS [15]	Two autoencoders combined with multiple classifiers and LightGBM-based fusion	Reconstruction errors from two autoencoders and outputs of multiple classifiers are fused by LightGBM; original multi-feature-set and fusion strategy retained; unspecified parameters selected using the validation set.
DLL-IDS [16]	Five-hidden-layer FNN combined with an LID-based adversarial-example detector	Five hidden layers; LID neighborhood size $k = 10$; SVM-based adversarial-example detector; original LID-based detection procedure retained.
RHAD [17]	Hierarchical VAE-based anomaly detector with shallow and in-depth detection stages	VAE architecture $(41, 30, 20, 30, 41)$; learning rate $= 0.001$; batch size $= 128$; reconstruction-error screening followed by feature-impact-based in-depth detection.
Apollon [18]	Multi-classifier defense with Multi-Armed Bandit selection and Thompson sampling	Candidate classifiers include MLP, NB, RF, DT, and LR; MLP hidden size $= 32$; RF estimators $= 100$; MLP maximum iterations $= 200$; original dynamic classifier-selection strategy retained.
CNN-LSTM [19]	Distributed CNN–LSTM intrusion-detection architecture implemented under Apache Spark	Original distributed CNN–LSTM architecture retained; unspecified training parameters selected according to validation performance.
Jin-IDS [20]	OBLR-based oversampling, LOF outlier detection, metric learning, genetic-algorithm-based feature selection, and LightGBM classification	SMOTE with OBLR for resampling; LOF outlier proportion $P = 0.36$; GA crossover probability $= 0.7$, mutation probability $= 0.03$, population size $= 20$; LightGBM used as the final classifier.

APPENDIX G
SENSITIVITY TO SIMULATED COMMUNICATION IMPAIRMENTS

The main experiments use vehicle positions from the Wyoming CV BSM dataset to determine distance-based V2V/V2I communication feasibility. We further examine the effects of communication delay, update loss, and varying wireless conditions on graph construction and intrusion detection. The original traffic-to-vehicle mapping, 10-s mobility windows, 200-m V2V radius, and 500-m RSU radius are retained. Each graph node remains an individual network traffic record.

For each distance-feasible V2V vehicle pair or V2I vehicle–RSU link ℓ in window t, we simulate the availability of one effective communication update. Records mapped to the same vehicle–time-window pair share the relevant simulated link state. If an update is lost or arrives after the specified deadline, candidate relationships that depend on that link are excluded for the current window. No traffic-record nodes, features, or labels are removed.

Each feasible link has a good, moderate, or poor channel state $q_{\ell,t}$. Its state persists into the next 10-s window with probability 0.8; otherwise, it changes to either of the other two states with equal probability. The state of a newly feasible link is initialized uniformly. Table VII specifies the base update-loss probability p_q and the delay distribution for each state. These are controlled simulation settings; the Wyoming BSM records provide vehicle mobility states rather than measured V2V/V2I reception or delay.

TABLE VII. Simulated Link Impairment Settings

Channel state	p_q	$\delta_{50,q}$ (ms)	$\delta_{95,q}$ (ms)
Good	0.02	20	60
Moderate	0.10	60	180
Poor	0.25	120	360

For each feasible link, the update-loss probability is adjusted according to its normalized distance:

$$p_{\ell,t} = \min\left\{0.95,\ p_{q_{\ell,t}} + 0.10\left(\frac{d_{\ell,t}}{R_\ell}\right)^2\right\}, \tag{1}$$

where $d_{\ell,t}$ is the vehicle–vehicle or vehicle–RSU distance, and R_ℓ is the corresponding V2V or RSU radius. The reception indicator is sampled as $Z_{\ell,t} \sim \text{Bernoulli}(1 - p_{\ell,t})$. For a received update, its delay $\delta_{\ell,t}$ follows a log-normal distribution with the median and 95th percentile in Table VII; the log-scale standard deviation is $\sigma_q = \ln(\delta_{95,q}/\delta_{50,q})/1.645$.

Let $M^{\text{dist}}_{\ell,t}$ indicate distance-based feasibility. The availability of a link after applying reception and delay is

$$M_{\ell,t} = M^{\text{dist}}_{\ell,t} Z_{\ell,t} \mathbf{1}[\delta_{\ell,t} \le \delta_{\max}], \tag{2}$$

where $\delta_{\max}$ is the timely-arrival deadline. A link with $Z_{\ell,t} = 0$ is unavailable without sampling its delay. The resulting mask constrains candidate connections among traffic-record nodes; LSH-based neighbor retrieval, structural entropy-guided neighborhood selection, and LLR-based edge weighting then follow the original graph-construction procedure.

We compare distance-only feasibility with delay-only, loss-only, and combined delay-and-loss settings. The default dead-

TABLE VIII. Performance Evaluation under Simulated Communication Impairments

Dataset	Condition	Retained links (%)	Clean F1 / AUC	FGSM F1 / AUC	MIM F1 / AUC	CW F1 / AUC
NSL-KDD	Distance only	100.0	0.9820 ± 0.0016 0.9820 ± 0.0014	0.9770 ± 0.0021 0.9765 ± 0.0018	0.9775 ± 0.0023 0.9773 ± 0.0020	0.9064 ± 0.0047 0.9212 ± 0.0041
	Delay only, $\delta_{\max} = 100$ ms	72.0 ± 1.1	0.9730 ± 0.0025 0.9750 ± 0.0023	0.9670 ± 0.0031 0.9690 ± 0.0029	0.9672 ± 0.0034 0.9692 ± 0.0031	0.8920 ± 0.0057 0.9070 ± 0.0051
	Loss only	84.2 ± 0.9	0.9780 ± 0.0021 0.9792 ± 0.0019	0.9730 ± 0.0027 0.9738 ± 0.0025	0.9732 ± 0.0029 0.9740 ± 0.0027	0.9000 ± 0.0052 0.9150 ± 0.0047
	Delay + loss, $\delta_{\max} = 100$ ms	63.1 ± 1.2	0.9686 ± 0.0029 0.9716 ± 0.0026	0.9621 ± 0.0036 0.9650 ± 0.0032	0.9623 ± 0.0038 0.9652 ± 0.0034	0.8851 ± 0.0063 0.9004 ± 0.0057
	Delay + loss, $\delta_{\max} = 50$ ms	42.5 ± 1.4	0.9548 ± 0.0039 0.9609 ± 0.0035	0.9467 ± 0.0047 0.9523 ± 0.0042	0.9471 ± 0.0049 0.9526 ± 0.0044	0.8702 ± 0.0076 0.8871 ± 0.0068
	Delay + loss, $\delta_{\max} = 200$ ms	78.2 ± 1.0	0.9758 ± 0.0023 0.9772 ± 0.0021	0.9703 ± 0.0030 0.9715 ± 0.0027	0.9705 ± 0.0032 0.9718 ± 0.0029	0.8968 ± 0.0055 0.9119 ± 0.0049
UNSW-NB15	Distance only	100.0	0.9398 ± 0.0028 0.9336 ± 0.0025	0.9310 ± 0.0032 0.9301 ± 0.0029	0.9307 ± 0.0035 0.9285 ± 0.0031	0.8890 ± 0.0056 0.8903 ± 0.0051
	Delay only, $\delta_{\max} = 100$ ms	72.0 ± 1.1	0.9310 ± 0.0034 0.9255 ± 0.0032	0.9207 ± 0.0040 0.9198 ± 0.0038	0.9204 ± 0.0042 0.9182 ± 0.0040	0.8770 ± 0.0062 0.8782 ± 0.0059
	Loss only	84.2 ± 0.9	0.9356 ± 0.0032 0.9300 ± 0.0030	0.9266 ± 0.0038 0.9258 ± 0.0035	0.9263 ± 0.0040 0.9243 ± 0.0038	0.8835 ± 0.0059 0.8850 ± 0.0055
	Delay + loss, $\delta_{\max} = 100$ ms	63.1 ± 1.2	0.9264 ± 0.0039 0.9209 ± 0.0037	0.9153 ± 0.0047 0.9147 ± 0.0043	0.9150 ± 0.0049 0.9132 ± 0.0046	0.8701 ± 0.0069 0.8714 ± 0.0065
	Delay + loss, $\delta_{\max} = 50$ ms	42.5 ± 1.4	0.9126 ± 0.0051 0.9081 ± 0.0047	0.9000 ± 0.0060 0.8996 ± 0.0056	0.8997 ± 0.0062 0.8980 ± 0.0059	0.8541 ± 0.0080 0.8570 ± 0.0075
	Delay + loss, $\delta_{\max} = 200$ ms	78.2 ± 1.0	0.9337 ± 0.0034 0.9279 ± 0.0032	0.9240 ± 0.0041 0.9229 ± 0.0038	0.9237 ± 0.0043 0.9214 ± 0.0040	0.8802 ± 0.0061 0.8818 ± 0.0057

line is $\delta_{\max} = 100$ ms, with additional combined-setting evaluations at 50 and 200 ms. The 10-s window specifies when vehicle states and candidate relationships are updated; $\delta_{\max}$ specifies whether a simulated update is timely for that window.

We reuse the five independently trained SE-RGCL models from Appendix D. For each model, an independently seeded realization of the simulated communication conditions is used to reconstruct the test graph. Model parameters, data partitions, traffic-record nodes, features, labels, and attack budgets remain fixed across conditions; no retraining is performed. FGSM, MIM, and CW adversarial examples are regenerated for each reconstructed graph using the attack configurations specified in Section V-B of the main manuscript. For each run, link retention is the number of distance-feasible V2V/V2I link–window instances remaining available after communication screening divided by the total number of distance-feasible link–window instances. Table VIII reports the mean $\pm$ standard deviation of link retention, F1, and AUC over the five runs; each performance cell lists F1 above AUC.

Table VIII shows that simulated delay and update loss reduce the availability of distance-feasible V2V/V2I link–window instances, while SE-RGCL maintains favorable detection performance. Under the combined setting with $\delta_{\max} = 100$ ms, link retention is 63.1%, and clean F1 scores are 0.9686 and 0.9264 on NSL-KDD and UNSW-NB15, respectively, compared with 0.9820 and 0.9398 under distance-only feasibility. Tightening the deadline to 50 ms further reduces link availability and performance, whereas relaxing it to 200 ms brings the results closer to the distance-only setting. FGSM, MIM, and CW exhibit similar trends under the simulated communication conditions.

APPENDIX H
DETAILED ADVERSARIAL ROBUSTNESS RESULTS

This appendix reports the complete attack-strength results corresponding to the adversarial-attack settings in Section V-B of the main manuscript. FGSM, PGD, BIM, and MIM are evaluated at five perturbation budgets; CW is evaluated at five values of κ, while DeepFool is reported under its specified adaptive setting. Tables IX and X present Accuracy, Precision, Recall, F1 score, and AUC on NSL-KDD and UNSW-NB15, respectively.

TABLE IX
Adversarial Robustness of SE-RGCL Across Attack Settings on NSL-KDD

Attack	Attack Parameter	Acc	Precision	Recall	F1 score	AUC
FGSM	ε=0.01	0.9816	0.9818	0.9812	0.9815	0.9813
	ε=0.025	0.9811	0.9813	0.9810	0.9810	0.9808
	ε=0.05	0.9797	0.9796	0.9770	0.9794	0.9791
	ε=0.075	0.9785	0.9783	0.9774	0.9781	0.9779
	ε=0.1	0.9765	0.9767	0.9762	0.9772	0.9763
PGD	ε=0.01	0.9802	0.9808	0.9801	0.9805	0.9803
	ε=0.025	0.9753	0.9783	0.9777	0.9780	0.9778
	ε=0.05	0.9716	0.9724	0.9709	0.9720	0.9718
	ε=0.075	0.9638	0.9655	0.9631	0.9650	0.9648
	ε=0.1	0.9585	0.9588	0.9573	0.9580	0.9582
BIM	ε=0.01	0.9805	0.9811	0.9806	0.9808	0.9806
	ε=0.025	0.9785	0.9788	0.9771	0.9789	0.9783
	ε=0.05	0.9744	0.9747	0.9741	0.9745	0.9743
	ε=0.075	0.9712	0.9714	0.9705	0.9710	0.9708
	ε=0.1	0.9675	0.9678	0.9638	0.9598	0.9672
MIM	ε=0.01	0.9818	0.9821	0.9815	0.9818	0.9820
	ε=0.025	0.9814	0.9816	0.9812	0.9813	0.9811
	ε=0.05	0.9802	0.9805	0.9780	0.9801	0.9799
	ε=0.075	0.9793	0.9795	0.9784	0.9792	0.9790
	ε=0.1	0.9778	0.9781	0.9764	0.9778	0.9776
Deepfool	-	0.9800	0.9802	0.9798	0.9800	0.9798
CW	κ=0.0	0.9808	0.9809	0.9807	0.9808	0.9807
	κ=0.5	0.9500	0.9503	0.9497	0.9503	0.9499
	κ=1.0	0.9522	0.9510	0.9520	0.9518	0.9532
	κ=2.0	0.9300	0.9303	0.9281	0.9275	0.9297
	κ=4.0	0.9150	0.9153	0.9109	0.9059	0.9147

For the four L_∞ attacks, F1 decreases as ϵ increases on both datasets. At $\epsilon = 0.1$, the lowest F1 among these attacks is 0.9580 on NSL-KDD and 0.9170 on UNSW-NB15, both under PGD. Under CW, F1 reaches 0.9059 and 0.8893 at $\kappa = 4$ on the two datasets, respectively. DeepFool yields F1 scores of

TABLE X
Adversarial Robustness of SE-RGCL Across Attack Settings on UNSW-NB15

Attack	Attack Parameter	Acc	Precision	Recall	F1 score	AUC
FGSM	ε=0.01	0.9468	0.9529	0.9277	0.9387	0.9285
	ε=0.025	0.9460	0.9519	0.9268	0.9378	0.9279
	ε=0.05	0.9442	0.9501	0.9247	0.9357	0.9258
	ε=0.075	0.9420	0.9477	0.9220	0.9332	0.9231
	ε=0.1	0.9405	0.9460	0.9203	0.9314	0.9212
PGD	ε=0.01	0.9440	0.9500	0.9250	0.9360	0.9261
	ε=0.025	0.9410	0.9470	0.9220	0.9330	0.9230
	ε=0.05	0.9370	0.9430	0.9180	0.9290	0.9188
	ε=0.075	0.9300	0.9360	0.9110	0.9220	0.9115
	ε=0.1	0.9250	0.9310	0.9060	0.9170	0.9069
BIM	ε=0.01	0.9465	0.9525	0.9275	0.9385	0.9283
	ε=0.025	0.9450	0.9510	0.9260	0.9370	0.9271
	ε=0.05	0.9435	0.9490	0.9240	0.9350	0.9249
	ε=0.075	0.9410	0.9465	0.9210	0.9325	0.9217
	ε=0.1	0.9390	0.9445	0.9190	0.9300	0.9196
MIM	ε=0.01	0.9468	0.9529	0.9277	0.9387	0.9286
	ε=0.025	0.9457	0.9515	0.9265	0.9375	0.9273
	ε=0.05	0.9439	0.9495	0.9245	0.9355	0.9254
	ε=0.075	0.9420	0.9475	0.9220	0.9330	0.9228
	ε=0.1	0.9403	0.9457	0.9200	0.9311	0.9205
Deepfool	-	0.9217	0.9126	0.8725	0.8873	0.8733
CW	κ=0	0.9454	0.9513	0.9261	0.9370	0.9270
	κ=0.5	0.9285	0.9346	0.9094	0.9203	0.9102
	κ=1.0	0.9125	0.9186	0.8934	0.9043	0.8941
	κ=2.0	0.9025	0.9086	0.8834	0.8943	0.8839
	κ=4.0	0.8975	0.9036	0.8784	0.8893	0.8780

0.9800 and 0.8873 under its specified setting. These results provide the per-setting detail for the adversarial evaluations summarized in the main manuscript.

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